

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers — the numbers and the contexts are the only things that differ. Work every problem with no notes, then check yourself against the key. **80 points total.**

Part A — Vocabulary (10 pts)

Write the letter of the best description next to each term.

- | | |
|------------------------------------|--|
| 1. Function C | (A) The set of every output the rule actually produces. |
| 2. Domain G | (B) Feeding the output of one rule into another: $f(g(x))$. |
| 3. Range A | (C) A rule that pairs each input with exactly one output. |
| 4. Zero of a function E | (D) The rule that gives back the question: if $f(a) = b$, it sends b to a . |
| 5. Concave up I | (E) An input whose output is 0. |
| 6. Average rate of change H | (F) The simplest member of a family, before any shift or stretch. |
| 7. Parent function F | (G) The set of every input the rule is allowed to take. |
| 8. Composition B | (H) The one steady rate that would give the same change in output over the same run. |
| 9. One-to-one J | (I) Bending upward — the rate of change is increasing. |
| 10. Inverse function D | (J) No two different inputs share an output. |

Part B — Multiple Choice (12 pts)

Circle the best answer, then write its letter in the blank at the right.

- The statement $f(3) = 7$ tells you that
 (A) $f(7) = 3$ (B) 7 is an input of f (C) $(3, 7)$ is on the graph of f (D) f is increasing at $x = 3$ **C**
- On an interval where f is decreasing and concave up, the outputs are
 (A) falling faster and faster (B) rising more and more slowly (C) changing at a constant rate (D) falling, but more and more slowly **D**
- A table of f has equally spaced inputs and **constant second** differences. The best model is
 (A) linear (B) quadratic (C) exponential (D) none of these **B**
- The graph of $g(x) = f(x + 3)$ is the graph of f shifted
 (A) 3 left (B) 3 right (C) 3 up (D) 3 down **A**
- If $f(x) = x^2$ and $g(x) = x - 1$, then $(f \circ g)(4)$ equals
 (A) 15 (B) 16 (C) 9 (D) 3 **C**
- A function has an inverse that is also a function exactly when the function is

Part C — Short Answer & Computation (40 pts, 5 each)

Show your work in the space provided.

1. Let $f(x) = 2x^2 - 5$.

$$f(3) = 2(3)^2 - 5 \\ = 13$$

$$f(-2) = 2(-2)^2 - 5 \\ = 3$$

$$2x^2 - 5 = 3$$

$$x^2 = 4$$

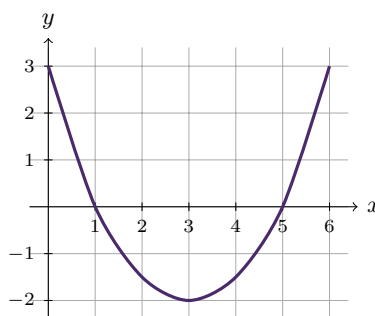
$$x = \pm 2$$

(a) $f(3) = 13$ (b) $f(-2) = 3$

(c) Solve $f(x) = 3$. There are two answers: $x = 2, -2$

(d) Which of (a)–(c) is a *solve* question rather than an *evaluate* question? (c)

2. The graph of $y = f(x)$ is shown below.



(a) On what interval is f decreasing? $0 < x < 3$

(b) Concave up or concave down on the whole picture? **Concave up**

(c) List the zeros of f : $x = 1$ and $x = 5$

(d) $f(3) = -2$ (e) Range on the window shown: $-2 \leq y \leq 3$

3. A tank is draining. Its volume V (gallons) after t minutes:

t (min)	0	4	8	12
V (gal)	180	148	124	108

$$\frac{V(8) - V(0)}{8 - 0} = \frac{124 - 180}{8} \\ = -7$$

(a) Average rate of change of V from $t = 0$ to $t = 8$: -7 , with units **gallons per minute**.

(b) In one sentence, what does that number mean for the tank?

Over those first 8 minutes the tank lost water at an average of 7 gallons a minute.

- (c) Is the tank draining faster or slower as time goes on? **Slower** Justify with the numbers.

The drop over each 4-minute block shrinks: -32 , then -24 , then -16 gallons......

4. The table below has equally spaced inputs. Fill in the difference rows, then name the function type.

x	0	1	2	3	4
$g(x)$	2	5	12	23	38
1st differences	—	3	7	11	15
2nd differences	—	—	4	4	4

Function type: **quadratic** Because **the 2nd differences are constant**

5. The parent function is $f(x) = \sqrt{x}$, with domain $[0, \infty)$ and range $[0, \infty)$. Let $g(x) = -2f(x - 1) + 4$.

- (a) Write g in terms of x : $g(x) = -2\sqrt{x-1} + 4$
 (b) The change *inside* the parentheses moves the graph **1 unit right**
 (c) The changes *outside*, in order: **stretch by 2, reflect over the x -axis, up 4**
 (d) Domain of g : **$[1, \infty)$** Range of g : **$(-\infty, 4]$**
 (e) The parent's landmark point $(0, 0)$ lands at **$(1, 4)$** on g .

6. Two one-to-one functions are given by the table below.

x	0	1	2	3	4
$f(x)$	5	2	7	1	3
$g(x)$	3	0	4	2	1

- (a) The handoff in $f(g(1))$ is **0**, so $f(g(1)) = \mathbf{5}$
 (b) $g(f(3)) = \mathbf{0}$
 (c) $f^{-1}(7) = \mathbf{2}$ because $f(\mathbf{2}) = 7$.
 (d) How can you tell from the table that f is one-to-one?
No output repeats in the f row, so no two inputs share an output......
 (e) Why does $g(f(0))$ have no value from this table?
 $f(0) = 5$, and 5 is not one of the inputs g accepts here......

7. Let $f(x) = 3x + 1$ and $g(x) = x^2 - 2$.

$$\begin{aligned}
 g(2) &= 4 - 2 \\
 &= 2 \\
 f(g(2)) &= 3(2) + 1 \\
 &= 7 \\
 f(2) &= 7 \\
 g(f(2)) &= 7^2 - 2 \\
 &= 47 \\
 (f \circ g)(x) &= 3(x^2 - 2) + 1 \\
 &= 3x^2 - 5
 \end{aligned}$$

- (a) $f(g(2)) = \mathbf{7}$ (b) $g(f(2)) = \mathbf{47}$

(c) $(f \circ g)(x) = 3x^2 - 5$

(d) What do (a) and (b) show about the order of a composition?

Order matters: $7 \neq 47$, so $f(g(x))$ and $g(f(x))$ are different functions......

8. Let $f(x) = \frac{x-5}{2}$.

$$y = \frac{x-5}{2}$$

$$x = \frac{y-5}{2}$$

$$2x = y - 5$$

$$y = 2x + 5$$

$$\begin{aligned} f(f^{-1}(x)) &= \frac{(2x+5)-5}{2} \\ &= x \end{aligned}$$

(a) Find f^{-1} by swapping and solving: $f^{-1}(x) = 2x + 5$

(b) Verify: $f(f^{-1}(x)) = x$

(c) $f(11) = 3$, so $f^{-1}(3) = 11$

Part D — Extended Response (18 pts, 9 each)

Explain your reasoning in full sentences.

1. A rectangular dog run is built against a barn wall, so only **three** sides need fencing. The builder has 60 feet of fence. Let x be the length of each of the two sides perpendicular to the barn.

$$\begin{aligned} A(x) &= x(60 - 2x) \\ &= 60x - 2x^2 \end{aligned}$$

$$\begin{aligned} A(5) &= 5(50) \\ &= 250 \end{aligned}$$

$$\begin{aligned} A(10) &= 10(40) \\ &= 400 \end{aligned}$$

$$\begin{aligned} \frac{A(10) - A(5)}{10 - 5} &= \frac{400 - 250}{5} \\ &= 30 \end{aligned}$$

(a) Which quantity is the *question* the model answers — its input? **x , the perpendicular side**

(b) The side parallel to the barn has length $60 - 2x$, so $A(x) = 60x - 2x^2$

(c) What type of function is A , and how do you know?

Quadratic: the highest power of x is 2, so second differences are constant......

(d) Domain the context forces: **$0 < x < 30$** Why?

A side must be positive, and the barn side $60 - 2x$ must be positive too......

(e) Average rate of change of A from $x = 5$ to $x = 10$: **30**, with units **ft² per foot of x** .

(f) State one assumption this model rests on.

No fence runs along the barn, and none is lost to a gate or an overlap......

2. Let $f(x) = x^2$ with domain all real numbers.

- (a) Explain why f has no inverse function on that domain. Give a counterexample pair of inputs.
Not one-to-one: $f(2) = f(-2) = 4$, so reversing sends 4 to two outputs.
- (b) Give a domain restriction that fixes the problem: $x \geq 0$ Then $f^{-1}(x) = \sqrt{x}$
- (c) Domain of that inverse: $[0, \infty)$ Range: $[0, \infty)$
- (d) Explain how the graph of a function and the graph of its inverse are related, and why.
They are reflections of each other over the line $y = x$
The inverse trades input for output, so (a, b) on f becomes (b, a) on f^{-1}